

MH1201 Linear Algebra II

Tutorial 4 (Week 5) – Problems & Solutions

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Overview

This tutorial focuses on core MH1201 themes:

- verifying linearity of a given map by definition and by “test vectors”;
- structure of linear maps on one-dimensional spaces (every endomorphism is a scalar multiple of the identity);
- how (non-)injectivity affects linear dependence under images;
- constructing a basis for $\mathcal{L}(V, W)$ and proving $\dim(\mathcal{L}(V, W)) = (\dim V)(\dim W)$;
- non-commutativity of linear operators when $\dim V > 1$;
- matrix representations under suitable choices of ordered bases (derivative operator);
- symmetric / skew-symmetric decompositions, rank-nullity, and standard matrices of linear maps on matrix spaces.

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Question 1 (Linearity of a map $\mathbb{R}^3 \rightarrow \mathbb{R}^2$)

Problem

Let $b, c \in \mathbb{R}$, and define a function $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by

$$\forall x, y, z \in \mathbb{R} : \quad T(x, y, z) := (2x - 4y + 3z + b, 6x + cxyz).$$

Show that $T \in \mathcal{L}(\mathbb{R}^3, \mathbb{R}^2)$ if and only if $b = c = 0$.

Solution

Method 1: Use the definition of linearity (zero and additivity tests)

Recall that T is linear if and only if:

$$T(\mathbf{0}) = \mathbf{0} \quad \text{and} \quad T(u + v) = T(u) + T(v) \quad \text{for all } u, v \in \mathbb{R}^3,$$

(or equivalently the full condition $T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$).

($\Rightarrow$) **Necessity.** First compute $T(0, 0, 0)$:

$$T(0, 0, 0) = (b, 0).$$

If T is linear then $T(0, 0, 0) = \mathbf{0}$, hence $b = 0$.

Next, still assuming linearity, use additivity with specific vectors. Let $u = (1, 0, 0)$ and $v = (0, 1, 1)$. Then $u + v = (1, 1, 1)$. Compute:

$$T(1, 1, 1) = (2 - 4 + 3 + b, 6 + c) = (1 + b, 6 + c).$$

Also

$$T(1, 0, 0) = (2 + b, 6), \quad T(0, 1, 1) = (-4 + 3 + b, 0) = (-1 + b, 0).$$

So

$$T(1, 0, 0) + T(0, 1, 1) = ((2 + b) + (-1 + b), 6 + 0) = (1 + 2b, 6).$$

By linearity, $T(u + v) = T(u) + T(v)$, so

$$(1 + b, 6 + c) = (1 + 2b, 6).$$

Comparing components gives $b = 0$ (again) and $c = 0$.

($\Leftarrow$) **Sufficiency.** If $b = c = 0$, then

$$T(x, y, z) = (2x - 4y + 3z, 6x).$$

This is of the form $T(x, y, z) = A \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ with

$$A = \begin{bmatrix} 2 & -4 & 3 \\ 6 & 0 & 0 \end{bmatrix},$$

hence T is linear.

$$\boxed{T \in \mathcal{L}(\mathbb{R}^3, \mathbb{R}^2) \iff b = c = 0}$$

Method 2: Homogeneity forces $b = 0$ and kills the nonlinear term xyz

Assume T is linear. Then for all $(x, y, z) \in \mathbb{R}^3$ and $\alpha \in \mathbb{R}$,

$$T(\alpha x, \alpha y, \alpha z) = \alpha T(x, y, z).$$

Step 1: $b = 0$ from $\alpha = 0$. Taking $\alpha = 0$ yields $T(0, 0, 0) = 0 \cdot T(x, y, z) = \mathbf{0}$, so $(b, 0) = \mathbf{0}$ and hence $b = 0$.

Step 2: $c = 0$ by comparing scaling of the second component. Now set $x = y = z = 1$. Then

$$T(1, 1, 1) = (2 - 4 + 3 + b, 6 + c) = (1 + b, 6 + c).$$

With $\alpha = 2$,

$$T(2, 2, 2) = (4 - 8 + 6 + b, 12 + 8c) = (2 + b, 12 + 8c),$$

since $(2)(2)(2) = 8$. But linearity gives $T(2, 2, 2) = 2T(1, 1, 1)$, so

$$(2 + b, 12 + 8c) = 2(1 + b, 6 + c) = (2 + 2b, 12 + 2c).$$

Comparing the first component gives $b = 0$, and comparing the second component gives

$$12 + 8c = 12 + 2c \quad \Rightarrow \quad 6c = 0 \quad \Rightarrow \quad c = 0.$$

Conversely, if $b = c = 0$ we already saw T is represented by a matrix, hence linear.

$$\boxed{T \in \mathcal{L}(\mathbb{R}^3, \mathbb{R}^2) \iff b = c = 0}$$

Question 2 (Linear maps on a one-dimensional space)

Problem

Let (i) V be a one-dimensional vector space over a field $\mathbb{F}$, and (ii) $T \in \mathcal{L}(V)$. Show that there exists a unique scalar $\lambda \in \mathbb{F}$ such that, for all $v \in V$, we have

$$T(v) = \lambda \cdot v.$$

Remark: A linear transformation on a one-dimensional vector space is therefore just a scaling.

Solution

Method 1: Choose a nonzero basis vector and extend by linearity

Since $\dim V = 1$, pick any nonzero vector $v_0 \in V$. Then $\{v_0\}$ is a basis of V . In particular, every $v \in V$ can be written uniquely as

$$v = av_0 \quad \text{for a unique } a \in \mathbb{F}.$$

Existence. Because $T(v_0) \in V$ and $\{v_0\}$ is a basis, there exists a unique scalar $\lambda \in \mathbb{F}$ such that

$$T(v_0) = \lambda v_0.$$

Now let $v = av_0$ be arbitrary. By linearity,

$$T(v) = T(av_0) = aT(v_0) = a(\lambda v_0) = \lambda(av_0) = \lambda v.$$

So this λ works for all v .

Uniqueness. Suppose $\lambda, \mu \in \mathbb{F}$ both satisfy $T(v) = \lambda v$ and $T(v) = \mu v$ for all v . Apply both to $v_0 \neq 0$:

$$\lambda v_0 = T(v_0) = \mu v_0 \quad \Rightarrow \quad (\lambda - \mu)v_0 = 0.$$

Since $v_0 \neq 0$, we must have $\lambda - \mu = 0$, hence $\lambda = \mu$.

$\exists! \lambda \in \mathbb{F} \text{ such that } T(v) = \lambda v \quad \forall v \in V$

Method 2: Identify V with $\mathbb{F}$ via an isomorphism

Pick any nonzero $v_0 \in V$. Define $\phi : \mathbb{F} \rightarrow V$ by $\phi(a) = av_0$. Then ϕ is a linear isomorphism because it is linear, surjective (every $v \in V$ equals av_0), and injective (if $av_0 = 0$ then $a = 0$ since $v_0 \neq 0$).

Consider the conjugated map on $\mathbb{F}$:

$$S := \phi^{-1} \circ T \circ \phi : \mathbb{F} \rightarrow \mathbb{F}.$$

Then S is linear (composition of linear maps). But $\mathbb{F}$ as a vector space over itself is one-dimensional with basis $\{1\}$. Therefore S is determined by $S(1) = \lambda$ for some $\lambda \in \mathbb{F}$, and for any $a \in \mathbb{F}$,

$$S(a) = S(a \cdot 1) = aS(1) = a\lambda = \lambda a.$$

Now for any $v \in V$, write $v = \phi(a)$, then

$$T(v) = T(\phi(a)) = \phi(S(a)) = \phi(\lambda a) = \lambda \phi(a) = \lambda v.$$

Uniqueness follows as in Method 1 by evaluating at any nonzero v .

$$T = \lambda \text{Id}_V \text{ for a unique } \lambda \in \mathbb{F}$$

Question 3 (Images of linearly dependent sets)

Problem

Let

- (i) V and W be vector spaces over the same field,
- (ii) $T \in \mathcal{L}(V, W)$, and
- (iii) S be a linearly dependent subset of V .

Is it necessarily true that the image $T[S]$ of S under T is a linearly dependent subset of W ?

Solution

Method 1: Counterexample (answer: no, not necessarily)

Claim: It is *not* necessarily true. We give an explicit counterexample.

Let $V = \mathbb{R}^3$ and $W = \mathbb{R}^2$. Define $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by

$$T(x, y, z) = (x, y).$$

This is linear (it is the coordinate projection onto the first two coordinates).

Now define the subset

$$S = \{v_1, v_2, v_3, v_4\} \subset \mathbb{R}^3, \quad v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad v_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Step 1: S is linearly dependent in V . Compute

$$v_3 - v_1 - v_4 + v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The coefficients $(1, -1, -1, 1)$ are not all zero, so S is linearly dependent.

Step 2: $T[S]$ is linearly independent in W . Compute the images:

$$T(v_1) = (1, 0), \quad T(v_2) = (0, 1), \quad T(v_3) = (0, 0), \quad T(v_4) = (1, 1).$$

Hence, as a set,

$$T[S] = \{(1, 0), (0, 1), (0, 0), (1, 1)\}.$$

This set is linearly independent in $\mathbb{R}^2$, since

$$a(1, 0) + b(0, 1) = (0, 0) \Rightarrow (a, b) = (0, 0).$$

Therefore $T[S]$ need not be linearly dependent even if S is linearly dependent.

No: S dependent does not force $T[S]$ dependent.

Method 2: The statement becomes true under injectivity (use contrapositive)

While the answer is *no* in general, it is useful to record a sufficient condition.

Claim: If T is injective, then S linearly dependent $\Rightarrow T[S]$ linearly dependent.

Proof (contrapositive). Assume $T[S]$ is linearly independent. We show S must be linearly independent.

Take any finite distinct vectors $v_1, \dots, v_k \in S$ and scalars $a_1, \dots, a_k$ such that

$$a_1 v_1 + \dots + a_k v_k = 0.$$

Apply T and use linearity:

$$a_1 T(v_1) + \dots + a_k T(v_k) = T(0) = 0.$$

Since $T[S]$ is linearly independent, the vectors $T(v_1), \dots, T(v_k)$ are linearly independent, so $a_1 = \dots = a_k = 0$. Thus every such relation is trivial, hence S is linearly independent.

Therefore, if S were dependent, $T[S]$ could not be independent; equivalently $T[S]$ must be dependent.

This shows precisely why non-injectivity is where counterexamples can occur (as in Method 1).

Question 4 (A basis for $\mathcal{L}(V, W)$)

Problem

Let

- (i) $m, n \in \mathbb{N}$,
 - (ii) $\mathbb{F}$ be a field,
 - (iii) V be an m -dimensional vector space over $\mathbb{F}$ with basis $\{v_1, \dots, v_m\}$, and
 - (iv) W be an n -dimensional vector space over $\mathbb{F}$ with basis $\{w_1, \dots, w_n\}$.
- Given $(i, j) \in \{1, \dots, m\} \times \{1, \dots, n\}$, let $T_{i,j} \in \mathcal{L}(V, W)$ be uniquely specified by the rule

$$\forall k \in \{1, \dots, m\} : \quad T_{i,j}(v_k) = \begin{cases} w_j, & k = i, \\ 0_W, & k \neq i. \end{cases}$$

Show that $\{T_{i,j}\}_{i=1,\dots,m; j=1,\dots,n}$ is a basis for $\mathcal{L}(V, W)$ (hence, $\dim(\mathcal{L}(V, W)) = mn$).

Solution

Method 1: Show spanning and linear independence directly

Step 1: Spanning. Let $T \in \mathcal{L}(V, W)$ be arbitrary. For each basis vector $v_i \in V$, we can expand $T(v_i)$ in the basis of W :

$$T(v_i) = \sum_{j=1}^n a_{ji} w_j \quad (a_{ji} \in \mathbb{F}).$$

Define a linear combination of the $T_{i,j}$ by

$$\tilde{T} := \sum_{i=1}^m \sum_{j=1}^n a_{ji} T_{i,j}.$$

We claim $\tilde{T} = T$. It suffices to check on the basis $\{v_1, \dots, v_m\}$. Fix $k \in \{1, \dots, m\}$. Then using the defining property of $T_{i,j}$,

$$\tilde{T}(v_k) = \sum_{i=1}^m \sum_{j=1}^n a_{ji} T_{i,j}(v_k) = \sum_{i=1}^m \sum_{j=1}^n a_{ji} \begin{cases} w_j, & k = i, \\ 0, & k \neq i, \end{cases} = \sum_{j=1}^n a_{jk} w_j = T(v_k).$$

Hence $\tilde{T}$ and T agree on a basis, so $\tilde{T} = T$. Therefore $\{T_{i,j}\}$ spans $\mathcal{L}(V, W)$.

Step 2: Linear independence. Suppose

$$\sum_{i=1}^m \sum_{j=1}^n \alpha_{i,j} T_{i,j} = 0 \quad (\text{the zero map}).$$

Apply both sides to v_k :

$$0 = \left(\sum_{i=1}^m \sum_{j=1}^n \alpha_{i,j} T_{i,j} \right) (v_k) = \sum_{i=1}^m \sum_{j=1}^n \alpha_{i,j} T_{i,j}(v_k) = \sum_{j=1}^n \alpha_{k,j} w_j,$$

because $T_{i,j}(v_k) = 0$ unless $i = k$. Since $\{w_1, \dots, w_n\}$ is a basis, the representation is unique, hence

$$\alpha_{k,1} = \dots = \alpha_{k,n} = 0.$$

As k was arbitrary, all $\alpha_{i,j} = 0$, proving linear independence.

Thus $\{T_{i,j}\}$ is a basis, and it contains mn elements, so

$$\boxed{\dim(\mathcal{L}(V, W)) = mn}.$$

Method 2: Identify $\mathcal{L}(V, W)$ with a matrix space $M_{n \times m}(\mathbb{F})$

Fix ordered bases $\beta = (v_1, \dots, v_m)$ for V and $\gamma = (w_1, \dots, w_n)$ for W . Each $T \in \mathcal{L}(V, W)$ has a matrix representation $[T]_{\gamma}^{\beta} \in M_{n \times m}(\mathbb{F})$ whose i th column is the coordinate vector of $T(v_i)$ in the basis γ .

Define

$$\Phi : \mathcal{L}(V, W) \rightarrow M_{n \times m}(\mathbb{F}), \quad \Phi(T) = [T]_{\gamma}^{\beta}.$$

Standard linear algebra shows Φ is a linear isomorphism (it is linear and bijective).

Now compute $\Phi(T_{i,j})$. By definition, $T_{i,j}(v_i) = w_j$ and $T_{i,j}(v_k) = 0$ for $k \neq i$. Hence the i th column of $[T_{i,j}]_{\gamma}^{\beta}$ is e_j (the j th standard basis vector of $\mathbb{F}^n$), and all other columns are 0. Therefore $\Phi(T_{i,j})$ is exactly the matrix unit E_{ji} (a 1 in row j , column i , and 0 elsewhere).

But $\{E_{ji}\}_{1 \leq j \leq n, 1 \leq i \leq m}$ is the standard basis of $M_{n \times m}(\mathbb{F})$. Since Φ is an isomorphism, the preimage set $\{T_{i,j}\}$ must be a basis of $\mathcal{L}(V, W)$, and its size is mn . Thus

$$\boxed{\{T_{i,j}\} \text{ is a basis of } \mathcal{L}(V, W) \text{ and } \dim(\mathcal{L}(V, W)) = mn}.$$

Method 3: Dimension-count via an explicit isomorphism $\mathcal{L}(V, W) \cong W^m$

Define $\Psi : \mathcal{L}(V, W) \rightarrow W^m$ by

$$\Psi(T) = (T(v_1), \dots, T(v_m)).$$

This is linear. It is injective because if $T(v_i) = 0$ for all i , then T is zero on a basis, hence $T = 0$. It is surjective because given any $(u_1, \dots, u_m) \in W^m$, there exists a unique linear map T such that $T(v_i) = u_i$ (define $T(\sum a_i v_i) = \sum a_i u_i$ and check well-definedness). Hence Ψ is an isomorphism, so

$$\dim(\mathcal{L}(V, W)) = \dim(W^m) = m \dim(W) = mn.$$

Moreover, the maps $T_{i,j}$ correspond to tuples with w_j in the i th slot and 0 elsewhere, which form a basis of W^m .

Question 5 (Non-commuting linear maps exist when $\dim V > 1$)

Problem

Let V be a finite-dimensional vector space, with $\dim(V) > 1$. Show that there exist $S, T \in \mathcal{L}(V)$ such that $S \circ T \neq T \circ S$.

Solution

Method 1: Construct operators using a basis and test on one vector

Since $\dim V > 1$, we can choose two linearly independent vectors $e_1, e_2 \in V$ and extend them to a basis $\beta = (e_1, e_2, e_3, \dots, e_n)$ of V .

Define $T, S \in \mathcal{L}(V)$ on the basis vectors by

$$T(e_1) = e_2, \quad T(e_2) = 0, \quad T(e_k) = 0 \quad (k \geq 3),$$

and

$$S(e_1) = e_1, \quad S(e_2) = 0, \quad S(e_k) = 0 \quad (k \geq 3).$$

Extend both maps linearly to all of V .

Now compute on e_1 :

$$(S \circ T)(e_1) = S(T(e_1)) = S(e_2) = 0,$$

while

$$(T \circ S)(e_1) = T(S(e_1)) = T(e_1) = e_2.$$

Hence $(S \circ T)(e_1) \neq (T \circ S)(e_1)$, so $S \circ T \neq T \circ S$.

$$\boxed{\exists S, T \in \mathcal{L}(V) \text{ with } S \circ T \neq T \circ S \quad (\dim V > 1)}$$

Method 2: Embed 2×2 matrices into $\mathcal{L}(V)$

Pick a 2-dimensional subspace $U \subset V$ (possible since $\dim V > 1$), and choose an ordered basis (e_1, e_2) of U . Define linear maps on U with matrices

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad (\text{with respect to } (e_1, e_2)).$$

Compute:

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = B, \quad BA = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0.$$

So $AB \neq BA$ on U .

Extend these maps to all of V by choosing a direct sum decomposition $V = U \oplus U'$ and defining both maps to be 0 on U' . Then the extensions (still denoted S, T) satisfy $ST \neq TS$ on V because they already disagree on U .

$$\boxed{\mathcal{L}(V) \text{ is non-commutative when } \dim V > 1}$$

Method 3: Commutator argument using rank and kernels

Let e_1, e_2 be linearly independent. Let T be any linear map such that $T(e_1) = e_2$ and $T(e_2) = 0$ (extend linearly). Let S be any linear map such that $S(e_2) = 0$ but $S(e_1) = e_1$. Then $(ST)(e_1) = S(e_2) = 0$ while $(TS)(e_1) = T(e_1) = e_2 \neq 0$. Hence they do not commute. This shows non-commutativity can be forced by arranging incompatible kernel/image constraints.

Question 6 (Choose bases so that $[D]_{\gamma}^{\beta} = [I_3 \ 0]$)

Problem

Let D denote the derivative operator from $P_3(\mathbb{R})$ to $P_2(\mathbb{R})$. Find an ordered basis β for $P_3(\mathbb{R})$ and an ordered basis γ for $P_2(\mathbb{R})$ such that

$$[D]_{\gamma}^{\beta} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

Solution

Method 1: Direct construction by choosing antiderivatives

We want $[D]_{\gamma}^{\beta}$ to have the following meaning:

$$D(\beta_1) = \gamma_1, \quad D(\beta_2) = \gamma_2, \quad D(\beta_3) = \gamma_3, \quad D(\beta_4) = 0.$$

So choose any ordered basis γ of $P_2(\mathbb{R})$, then pick $\beta_1, \beta_2, \beta_3$ as antiderivatives of $\gamma_1, \gamma_2, \gamma_3$, and pick β_4 as a nonzero element of $\ker D$ (i.e. a nonzero constant polynomial).

A convenient choice is

$$\gamma = (\gamma_1, \gamma_2, \gamma_3) = (1, x, x^2),$$

which is an ordered basis of $P_2(\mathbb{R})$.

Now choose

$$\beta = (\beta_1, \beta_2, \beta_3, \beta_4) = \left(x, \frac{x^2}{2}, \frac{x^3}{3}, 1\right).$$

Then

$$D(x) = 1, \quad D\left(\frac{x^2}{2}\right) = x, \quad D\left(\frac{x^3}{3}\right) = x^2, \quad D(1) = 0.$$

Therefore, the coordinate vectors of $D(\beta_i)$ in the basis γ are

$$[D(\beta_1)]_{\gamma} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad [D(\beta_2)]_{\gamma} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad [D(\beta_3)]_{\gamma} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad [D(\beta_4)]_{\gamma} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Hence

$$[D]_{\gamma}^{\beta} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix},$$

as required.

$$\beta = \left(x, \frac{x^2}{2}, \frac{x^3}{3}, 1\right), \quad \gamma = (1, x, x^2)$$

Method 2: Kernel–image decomposition and basis adaptation

Let $V = P_3(\mathbb{R})$ and $W = P_2(\mathbb{R})$. The derivative map $D : V \rightarrow W$ is linear.

Step 1: Identify $\ker D$ and $\text{range } D$.

$$\ker D = \{p \in P_3(\mathbb{R}) : p'(x) = 0 \ \forall x\} = \{\text{constant polynomials}\} = \text{span}\{1\}.$$

Also, every polynomial in $P_2(\mathbb{R})$ has an antiderivative in $P_3(\mathbb{R})$, so D is surjective:

$$\text{range } D = P_2(\mathbb{R}).$$

Step 2: Choose bases compatible with an isomorphism on a complement. Pick any ordered basis $\gamma = (\gamma_1, \gamma_2, \gamma_3)$ of $W = P_2(\mathbb{R})$ (e.g. $\gamma = (1, x, x^2)$). For each γ_i , choose $\beta_i \in V$ such that $D(\beta_i) = \gamma_i$; this is possible since D is surjective. Finally, choose β_4 to be a basis vector for $\ker D$ (e.g. $\beta_4 = 1$).

Then the matrix $[D]_{\gamma}^{\beta}$ has columns equal to $[D(\beta_i)]_{\gamma}$; by construction, the first three columns are e_1, e_2, e_3 and the last is 0. Hence $[D]_{\gamma}^{\beta} = [I_3 \ 0]$.

A concrete choice recovered from this method is exactly Method 1: $\gamma = (1, x, x^2)$ and $\beta = (x, x^2/2, x^3/3, 1)$.

Method 3: Change-of-basis from the monomial representation

With respect to the monomial bases

$$\beta_{\text{mon}} = (1, x, x^2, x^3), \quad \gamma_{\text{mon}} = (1, x, x^2),$$

we have

$$D(1) = 0, \quad D(x) = 1, \quad D(x^2) = 2x, \quad D(x^3) = 3x^2,$$

so

$$[D]_{\gamma_{\text{mon}}}^{\beta_{\text{mon}}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

Now scale the domain basis vectors x^2, x^3 by $1/2, 1/3$ respectively and reorder the domain basis as $(x, x^2/2, x^3/3, 1)$ while keeping $\gamma = (1, x, x^2)$. This turns the above matrix into $[I_3 \ 0]$.

Question 7 (The map $T(M) = M - M^T$ on matrix spaces)

Problem

Letting $n \in \mathbb{N}$, define a function $T : \mathbb{R}^{n,n} \rightarrow \mathbb{R}^{n,n}$ by

$$\forall M \in \mathbb{R}^{n,n} : \quad T(M) := M - M^T.$$

- (a) Show that $T \in \mathcal{L}(\mathbb{R}^{n,n})$.
- (b) Find the standard matrix representation of T in the case $n = 2$.

Remark: The standard ordered basis for $\mathbb{R}^{2,2}$ is

$$\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right).$$

- (c) Find nullity(T).
- (d) Find rank(T).

Solution

Method 1: Direct computation (linearity, standard matrix, then rank-nullity)

- (a) **Linearity.** Let $A, B \in \mathbb{R}^{n,n}$ and $\alpha, \beta \in \mathbb{R}$. Using $(A + B)^T = A^T + B^T$ and $(\alpha A)^T = \alpha A^T$,

$$\begin{aligned} T(\alpha A + \beta B) &= (\alpha A + \beta B) - (\alpha A + \beta B)^T \\ &= \alpha A + \beta B - (\alpha A^T + \beta B^T) \\ &= \alpha(A - A^T) + \beta(B - B^T) \\ &= \alpha T(A) + \beta T(B). \end{aligned}$$

Hence $T \in \mathcal{L}(\mathbb{R}^{n,n})$.

- (b) **Standard matrix for $n = 2$.** Let the standard ordered basis be

$$(E_{11}, E_{12}, E_{21}, E_{22}) = \left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right).$$

Compute T on basis elements:

$$T(E_{11}) = E_{11} - E_{11}^T = 0, \quad T(E_{22}) = E_{22} - E_{22}^T = 0,$$

and

$$T(E_{12}) = E_{12} - E_{21}, \quad T(E_{21}) = E_{21} - E_{12} = -(E_{12} - E_{21}).$$

Write these in coordinates relative to $(E_{11}, E_{12}, E_{21}, E_{22})$:

$$[T(E_{11})] = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad [T(E_{12})] = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}, \quad [T(E_{21})] = \begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \quad [T(E_{22})] = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Therefore, the standard matrix of T for $n = 2$ is the 4×4 matrix with these vectors as columns:

$$[T] = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Thus

$$\boxed{[T]_{(E_{11}, E_{12}, E_{21}, E_{22})} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}}.$$

(c) $\text{nullity}(T)$. We have $T(M) = 0 \iff M - M^T = 0 \iff M = M^T$, i.e. M is symmetric. A real symmetric $n \times n$ matrix is determined by:

- n diagonal entries, and
- $\frac{n(n-1)}{2}$ entries above the diagonal (the entries below are forced by symmetry).

Hence

$$\text{nullity}(T) = \dim(\ker T) = n + \frac{n(n-1)}{2} = \frac{n(n+1)}{2}.$$

So

$$\boxed{\text{nullity}(T) = \frac{n(n+1)}{2}}.$$

(d) $\text{rank}(T)$. By rank-nullity on the vector space $\mathbb{R}^{n,n}$ of dimension n^2 ,

$$\text{rank}(T) = n^2 - \text{nullity}(T) = n^2 - \frac{n(n+1)}{2} = \frac{2n^2 - n^2 - n}{2} = \frac{n(n-1)}{2}.$$

Thus

$$\boxed{\text{rank}(T) = \frac{n(n-1)}{2}}.$$

Method 2: Conceptual decomposition into symmetric and skew-symmetric parts

Define the subspaces

$$\text{Sym}_n = \{M \in \mathbb{R}^{n,n} : M^T = M\}, \quad \text{Skew}_n = \{M \in \mathbb{R}^{n,n} : M^T = -M\}.$$

Step 1: Direct sum decomposition.

For any $M \in \mathbb{R}^{n,n}$,

$$M = \underbrace{\frac{M + M^T}{2}}_{\in \text{Sym}_n} + \underbrace{\frac{M - M^T}{2}}_{\in \text{Skew}_n}.$$

This decomposition is unique and

$$\mathbb{R}^{n,n} = \text{Sym}_n \oplus \text{Skew}_n.$$

(a) Linearity.

The transpose map is linear, hence

$$T(M) = M - M^T$$

is the difference of two linear maps, and therefore linear. Thus,

$$T \in \mathcal{L}(\mathbb{R}^{n,n}).$$

(b) Standard matrix for $n = 2$.

Using the standard ordered basis

$$(E_{11}, E_{12}, E_{21}, E_{22}),$$

we compute

$$\begin{aligned} T(E_{11}) &= 0, & T(E_{22}) &= 0, \\ T(E_{12}) &= E_{12} - E_{21}, & T(E_{21}) &= E_{21} - E_{12}. \end{aligned}$$

Hence the coordinate vectors are

$$[T(E_{11})] = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad [T(E_{12})] = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}, \quad [T(E_{21})] = \begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \quad [T(E_{22})] = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Thus

$$[T] = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

(c) Nullity.

We have

$$T(M) = 0 \iff M - M^T = 0 \iff M = M^T.$$

Hence

$$\ker T = \text{Sym}_n.$$

Counting parameters:

$$\dim(\text{Sym}_n) = n + \frac{n(n-1)}{2} = \frac{n(n+1)}{2}.$$

Therefore

$$\boxed{\text{nullity}(T) = \frac{n(n+1)}{2}}.$$

(d) Rank.

For any M ,

$$T(M)^T = (M - M^T)^T = M^T - M = -T(M),$$

so $T(M)$ is skew-symmetric. Thus

$$\text{range}(T) \subseteq \text{Skew}_n.$$

Conversely, if $K \in \text{Skew}_n$, let $M = \frac{1}{2}K$. Then

$$T(M) = K.$$

Hence

$$\text{range}(T) = \text{Skew}_n.$$

Counting parameters:

$$\dim(\text{Skew}_n) = \frac{n(n-1)}{2}.$$

Thus

$$\boxed{\text{rank}(T) = \frac{n(n-1)}{2}}.$$

Method 3: Specialization to $n = 2$ as a consistency check

Let

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Then

$$T(M) = M - M^T = \begin{bmatrix} 0 & b-c \\ c-b & 0 \end{bmatrix}.$$

(a) Linearity.

Since transpose is linear and subtraction preserves linearity,

$$\boxed{T \in \mathcal{L}(\mathbb{R}^{2,2})}.$$

(b) Standard matrix.

Using the ordered basis

$$(E_{11}, E_{12}, E_{21}, E_{22}),$$

the matrix representation is

$$[T] = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

(c) Nullity.

We have

$$T(M) = 0 \iff b = c.$$

Thus a, b, d are free parameters. Hence

$$\boxed{\text{nullity}(T) = 3}.$$

(d) Rank.

Since $\dim(\mathbb{R}^{2,2}) = 4$,

$$\text{rank}(T) = 4 - 3 = 1.$$

Therefore

$$\boxed{\text{rank}(T) = 1}.$$

These agree with the general formulas

$$\frac{n(n+1)}{2} \Big|_{n=2} = 3, \quad \frac{n(n-1)}{2} \Big|_{n=2} = 1.$$